Demographics

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Demographic characterization

```
metavars <- c("Residence","Age",  "Diet", "Contraceptives", "Regular/daily medication" , "Nic
metavars2 <- c("Coitus","Alcohol", "Medication during sampling")

metadata <- rio::import('~/Dropbox/Bekzod&Isabella/ME_translation_key/Metadata.xlsx', sheet =
  filter(!is.na(Residence))
colnames(metadata)[colnames(metadata)=='Nicotin'] <- 'Nicotine'

colnames(MEconc_newlong)[colnames(MEconc_newlong)=='Nicotin'] <- 'Nicotine'
colnames(PBconc_newlong)[colnames(PBconc_newlong)=='Nicotin'] <- 'Nicotine'


colnames(phyX@sam_data)[colnames(phyX@sam_data)=='Nicotin'] <- 'Nicotine'
colnames(phyITS@sam_data)[colnames(phyITS@sam_data)=='Nicotin'] <- 'Nicotine'

dfmeta <- phyX@sam_data %>%
  data.frame() %>%
  filter(!is.na(Participant))
```

```r
hasdata <- full_join(
  data.frame(Sample.name = paste(dfmeta$Participant,dfmeta$cycle,'.',dfmeta$day, sep = ''),
             has_seq = 1),
  data.frame(Sample.name = paste(MEconc_newlong$Participant,MEconc_newlong$Cycle,'.',MEconc_
  by = 'Sample.name') %>%
  mutate(hasdata = ifelse(has_seq==1 | has_nmr==1,1,0))

dftb1 <- metadata %>%
  filter(`Blood origen`=='ME') %>%
  left_join(hasdata, by = c('Sample ID' = 'Sample.name')) %>%
  group_by(Participant) %>%
  mutate(n_seq = sum(has_seq, na.rm = T),
         n_NMR = sum(has_nmr, na.rm = T),
         Coitus_any = (sum(Coitus,na.rm = T)>0) %>% as.numeric(),
         Alcohol_any = (sum(Alcohol,na.rm = T)>0) %>% as.numeric(),
         Meds_any = (sum(`Medication during sampling`,na.rm = T)>0) %>% as.numeric()) %>%
  filter(!duplicated(Participant)) %>%
  mutate(missNMR = ifelse(n_NMR > 6,'>median_nobs','<median_nobs'),
         missSeq = ifelse(n_seq ==9,'>median_nobs','<median_nobs'))

tb1 <- dftb1 %>% tableone::CreateTableOne(data = ., vars = c(metavars,'n_seq','n_NMR','Coitus

tb1missNMR <- dftb1 %>% tableone::CreateTableOne(data = .,
                                                 strata = 'missNMR',
                                                 vars = c(metavars,'n_seq','Coitus_any','Alcohol

tb1missSeq <- dftb1 %>% tableone::CreateTableOne(data = .,
                                                 strata = 'missSeq',
                                                 vars = c(metavars,'n_NMR','Coitus_any','Alcohol


tab1AMat <- print(tb1, nonnormal = c('n_seq','n_NMR'), printToggle = F)

tab1AMat_NMR <- print(tb1missNMR, nonnormal = c('n_seq','n_NMR'), printToggle = F)
tab1AMat_SEQ <- print(tb1missSeq, nonnormal = c('n_seq','n_NMR'), printToggle = F)


tb12 <- metadata %>%
  filter(`Blood origen`=='ME') %>%
  left_join(hasdata, by = c('Sample ID' = 'Sample.name')) %>%
  filter(hasdata == 1) %>%
  tableone::CreateTableOne(data = ., vars = metavars2, factorVars = metavars2)
```

```
tab1BMat <- print(tb12, nonnormal = c('n_seq','n_NMR'), printToggle = F)

tab1AMat %>% knitr::kable(caption = 'Participant level demographic characterization')
```

Table 1: Participant level demographic characterization

	Overall
n	19
Residence = C (%)	8 (42.1)
Age (%)	
M	8 (42.1)
O	6 (31.6)
Y	5 (26.3)
Diet (%)	
F	6 (31.6)
O	8 (42.1)
V	5 (26.3)
Contraceptives = 1 (%)	3 (15.8)
Regular/daily medication = Y (%)	5 (26.3)
Nicotine = 1 (%)	6 (31.6)
n_seq (median [IQR])	9.00 [8.00, 9.00]
n_NMR (median [IQR])	6.00 [4.50, 8.00]
Coitus_any = 1 (%)	7 (36.8)
Alcohol_any = 1 (%)	16 (84.2)
Meds_any = 1 (%)	9 (47.4)

```
tab1AMat_NMR %>% knitr::kable(caption = 'Participant level demographic characterization - st
```

Table 2: Participant level demographic characterization - stratified by number of observed samples with NMR data

	<median_nobs	>median_nobs	p	test
n	11	8		
Residence = C (%)	5 (45.5)	3 (37.5)	1.000	
Age (%)			0.890	
M	5 (45.5)	3 (37.5)		
O	3 (27.3)	3 (37.5)		
Y	3 (27.3)	2 (25.0)		
Diet (%)			0.278	

	<median_nobs	>median_nobs	p	test
F	2 (18.2)	4 (50.0)		
O	5 (45.5)	3 (37.5)		
V	4 (36.4)	1 (12.5)		
Contraceptives = 1 (%)	2 (18.2)	1 (12.5)	1.000	
Regular/daily medication = Y (%)	3 (27.3)	2 (25.0)	1.000	
Nicotine = 1 (%)	3 (27.3)	3 (37.5)	1.000	
n_seq (median [IQR])	8.00 [6.50, 9.00]	9.00 [8.00, 9.00]	0.344	non-norm
Coitus_any = 1 (%)	4 (36.4)	3 (37.5)	1.000	
Alcohol_any = 1 (%)	8 (72.7)	8 (100.0)	0.331	
Meds_any = 1 (%)	5 (45.5)	4 (50.0)	1.000	

```
tab1AMat_SEQ %>% knitr::kable(caption = 'Participant level demographic characterization - str
```

Table 3: Participant level demographic characterization - stratified by number of observed samples with Sequencing data

	<median_nobs	>median_nobs	p	test
n	9	10		
Residence = C (%)	4 (44.4)	4 (40.0)	1.000	
Age (%)			0.324	
M	5 (55.6)	3 (30.0)		
O	3 (33.3)	3 (30.0)		
Y	1 (11.1)	4 (40.0)		
Diet (%)			0.723	
F	3 (33.3)	3 (30.0)		
O	3 (33.3)	5 (50.0)		
V	3 (33.3)	2 (20.0)		
Contraceptives = 1 (%)	2 (22.2)	1 (10.0)	0.921	
Regular/daily medication = Y (%)	1 (11.1)	4 (40.0)	0.365	
Nicotine = 1 (%)	3 (33.3)	3 (30.0)	1.000	
n_NMR (median [IQR])	5.00 [3.00, 7.00]	7.00 [5.25, 8.00]	0.064	non-norm
Coitus_any = 1 (%)	3 (33.3)	4 (40.0)	1.000	
Alcohol_any = 1 (%)	8 (88.9)	8 (80.0)	1.000	
Meds_any = 1 (%)	4 (44.4)	5 (50.0)	1.000	

```
tab1BMat %>% knitr::kable(caption = 'Sample level demographic characterization')
```

Table 4: Sample level demographic characterization

	Overall
n	151
Coitus = 1 (%)	12 (8.2)
Alcohol = 1 (%)	51 (34.9)
Medication during sampling = 1 (%)	17 (38.6)

```
library(openxlsx)
wb <- createWorkbook()

addWorksheet(wb, "Participant")
tab1AMat %>% as.data.frame() %>% rownames_to_column() %>%
writeData(wb, "Participant", x = .)
addWorksheet(wb, "Sample")
tab1BMat %>% as.data.frame() %>% rownames_to_column() %>%
writeData(wb, "Sample", x = .)
# Save file
# saveWorkbook(wb, "TableOne.xlsx", overwrite = TRUE)
```

Betadiversity

Effect of demographics / meta-information

```
metavars <- c("Age", "Residence", "Diet", "Contraceptives", "Regular/daily medication" , "Ni
dfmeta <- MEconc_newlong %>%
  filter(!duplicated(`Samples #`)) %>%
  dplyr::select(`Samples #`,Participant,`Cycle day`,metavars) %>%
  gather(demo,val,-`Samples #`,-Participant,-`Cycle day`)

dfmetaPB <- PBconc_newlong %>%
  filter(!duplicated(`Samples #`)) %>%
  dplyr::select(`Samples #`,Participant,metavars) %>%
  gather(demo,val,-`Samples #`,-Participant)

SCORESdemo <- resDEMO <- data.frame()
nperm <- 9999
```

```r
for (i in  1:length(metavars)){
  # for (i in  1:2){
  # i <- 3
  mvar <- metavars[i]
  mvar2 <- mvar %>% gsub(' ','.',x = .) %>% gsub('/','.',x = .)
  print(mvar2)
  ######## ME-NMR
  dfsel <- dfmeta %>%
    filter(demo==mvar) %>%
    filter(!is.na(val)) %>%
    arrange()

  dfselun <- dfsel %>% ungroup %>%  filter(!duplicated(Participant))

  df <- MEconc_newlong %>%
    filter(!is.na(Participant)) %>%
    filter(`Samples #` %in% dfsel$`Samples #`) %>%
    filter(!str_detect(met,'PC')) %>%
    filter(!(str_detect(met,'glycerol') | met == 'ERECTIC' | met=='TSP' | str_detect(met,'PC
    filter(!(str_detect(met,'glycerol') | str_detect(met,'EDTA') | met == 'ERECTIC' | met=='T
    dplyr::select(`Samples #`, met,logy) %>%
    spread(met,logy)

  rME <- adonis2(scale(df) ~ factor(`Cycle day`) + val, data=dfsel, permutations=0, method =
    tidy %>%
    mutate(demo = mvar, omics = 'ME-NMR')

  rMEperm <- data.frame()
  for (j in 1:nperm){
    dfselperm <- dfsel %>%
      left_join(dfselun %>% dplyr::select(-`Cycle day`) %>%  mutate(val2 = sample(val)), by =
    rMEperm <- rMEperm %>%
      bind_rows(adonis2(scale(df) ~ factor(`Cycle day`) + val2, data=dfselperm,
                        permutations=0, method = 'euclidean', by = 'terms') %>%
                  tidy %>% filter(term=='val2') %>%
                  mutate(demo = mvar, omics = 'ME-NMR', type = 'perm') )
  }

  pv <- sum(rMEperm$statistic>rME$statistic[2]) / dim(rMEperm)[1]
  rME$p.value2 <- NA
  rME$p.value2[2] <- pv
```

```r
### vectors for PCA
mPCA_i <- prcomp(df, scale. = T)
scrsME <- mPCA_i$x[,1:4] %>%
  data.frame() %>%
  mutate(demo = mvar,
         val = dfsel$val %>% as.character(),
         omics = 'ME-NMR')


######## PB-NMR
dfsel <- dfmetaPB %>%
  filter(demo==mvar) %>%
  filter(!is.na(val)) %>%
  arrange()

dfselun <- dfsel %>% ungroup %>%  filter(!duplicated(Participant))

if ((dfsel$val %>% unique() %>% length())==1 | dim(dfsel)[1]==0) {
  print(i)
  rPB <- data.frame(term='Model',R2 = NA, p.value = 1) %>% mutate(demo = mvar, omics = 'PB-
}
else {

  df <- PBconc_newlong %>%
    filter(!is.na(Participant)) %>%
    filter(`Samples #` %in% dfsel$`Samples #`) %>%
    filter(!str_detect(met,'PC')) %>%
    filter(!(str_detect(met,'glycerol') | met == 'ERECTIC' | met=='TSP' | str_detect(met,'
    filter(!(str_detect(met,'glycerol') | str_detect(met,'EDTA') | met == 'ERECTIC' | met==
    dplyr::select(`Samples #`, met,logy) %>%
    spread(met,logy)

  rPB <- adonis2(scale(df) ~ val, data=dfsel, permutations=0, method = 'euclidean')  %>% ti
    mutate(demo = mvar, omics = 'PB-NMR')

  rPBperm <- data.frame()
  for (j in 1:nperm){
    dfselperm <- dfsel %>%
      left_join(dfselun %>% mutate(val2 = sample(val)), by = 'Participant')
    rPBperm <- rPBperm %>%
      bind_rows(adonis2(scale(df) ~ val2, data=dfselperm,
                        permutations=0, method = 'euclidean') %>%
```

```
                    tidy %>% filter(term=='Model') %>%
                    mutate(demo = mvar, omics = 'ME-NMR', type = 'perm') )
  }

  pv <- sum(rPBperm$statistic>rPB$statistic[1]) / dim(rPBperm)[1]
  rPB$p.value2 <- NA
  rPB$p.value2[1] <- pv

  ### vectors for PCA
  mPCA_i <- prcomp(df, scale. = T)
  scrsPB <- mPCA_i$x[,1:4] %>%
    data.frame() %>%
    mutate(demo = mvar,
           val = dfsel$val %>% as.character(),
           omics = 'ME-NMR')
}

# phyX@sam_data$demo <- phyX@sam_data %>% data.frame() %>% dplyr::select(mvar2)

####### 16s
phyXsel <- phyX
samdf <- data.frame(phyX@sam_data,demo = phyX@sam_data %>% data.frame() %>% dplyr::select(m
phyXsel@sam_data <- sample_data(samdf)
phyXsel <- phyXsel %>%
  subset_samples(!is.na(demo)) %>%
  transform_sample_counts( function(x) x  / sum(x))
D16s <- phyXsel %>% phyloseq::distance("bray")

dfsel <- data.frame(sample_data(phyXsel))
dfselun <- dfsel%>% filter(!duplicated(Participant))

r16s <- adonis2(D16s ~ demo, data = dfsel, permutations = 0) %>% tidy %>%
  mutate(demo = mvar, omics = 'ME-16s')

r16sperm <- data.frame()
for (j in 1:nperm){
  dfselperm <- dfsel %>%
    left_join(dfselun %>% mutate(demo2 = sample(demo)), by = 'Participant')
  r16sperm <- r16sperm %>%
    bind_rows(adonis2(D16s ~ demo2, data = dfselperm, permutations = 0) %>% tidy %>%
                mutate(demo = mvar, omics = 'ME-16s') %>% filter(term=='Model'))
}
```

```r
pv <- sum(r16sperm$statistic>r16s$statistic[1]) / dim(r16sperm)[1]
r16s$p.value2 <- NA
r16s$p.value2[1] <- pv


### vectors for PCoA
NMDS <- monoMDS(D16s, k = 4)
scrsME16s <- NMDS$points[,1:4] %>% data.frame()
colnames(scrsME16s) <- paste('PC',1:4, sep = '')
scrsME16s <- scrsME16s %>%
  mutate(demo = mvar,
         val = dfsel$demo %>% as.character(),
         omics = 'ME-16s')

##### ITS
phyXsel <- phyITS
samdf <- data.frame(phyITS@sam_data,demo = phyITS@sam_data %>% data.frame() %>% dplyr::sele
phyXsel@sam_data <- sample_data(samdf)
phyXsel <- phyXsel %>%
  subset_samples(!is.na(demo)) %>%
  transform_sample_counts( function(x) as.numeric(x>0))
DITS <- phyXsel %>% phyloseq::distance("euclidean")

dfsel <- data.frame(sample_data(phyXsel))
dfselun <- dfsel%>% filter(!duplicated(Participant))

rITS <- adonis2(DITS ~ demo, data = dfsel, permutations = 0) %>% tidy %>%
  mutate(demo = mvar, omics = 'ME-ITS')

rITSperm <- data.frame()
for (j in 1:nperm){
  dfselperm <- dfsel %>%
    left_join(dfselun %>% mutate(demo2 = sample(demo)), by = 'Participant')
  rITSperm <- rITSperm %>%
    bind_rows(adonis2(DITS ~ demo2, data = dfselperm, permutations = 0) %>% tidy %>%
                mutate(demo = mvar, omics = 'ME-ITS') %>% filter(term=='Model'))
}

pv <- sum(rITSperm$statistic>rITS$statistic[1]) / dim(rITSperm)[1]
rITS$p.value2 <- NA
rITS$p.value2[1] <- pv
```

```
  ### vectors for PCoA
  NMDS <- monoMDS(DITS, k = 4)
  scrsMEITS <- NMDS$points[,1:4] %>% data.frame()
  colnames(scrsMEITS) <- paste('PC',1:4, sep = '')

  scrsMEITS <- scrsMEITS %>%
    mutate(demo = mvar,
           val = dfsel$demo %>% as.character(),
           omics = 'ME-ITS')

  phyXsel <- phyITS
  samdf <- data.frame(phyITS@sam_data,demo = phyITS@sam_data %>% data.frame() %>% dplyr::sele
  phyXsel@sam_data <- sample_data(samdf)
  phyXsel <- phyXsel %>%
    subset_samples(!is.na(demo)) %>%
    transform_sample_counts( function(x) sqrt(x))
  DITS <- phyXsel %>% phyloseq::distance("euclidean")
  rITS2 <- adonis2(DITS ~ demo, data = data.frame(sample_data(phyXsel))) %>%
    tidy %>%
    mutate(demo = mvar, omics = 'ME-ITS-sqrt_counts')


  resDEMO <- bind_rows(resDEMO,rME,rPB,r16s,rITS,rITS2)

  scrs <- bind_rows(scrsME,scrsME16s,scrsMEITS,scrsPB)

  SCORESdemo <- bind_rows(SCORESdemo,scrs)

}

save(file = 'betademo_res.RData', list = c('SCORESdemo','resDEMO','metavars'))
```

Robustness of betadiversity associations

We test whether rarefaction and sample-removal causes 16s and ITS results to cause differential results.

```
phyXrare2k <- rarefy_even_depth(phyX,2000)
phyXrare10k <- rarefy_even_depth(phyX,10000)
phyITS_100 <- prune_samples(sample_sums(phyITS) >= 100,phyITS)
phyITS_00 <- prune_samples(sample_sums(phyITS) > 0,phyITS)
```

```r
metavars <- c("Age", "Residence", "Diet", "Contraceptives", "Regular/daily medication" , "Ni

resDEMO_stab <- data.frame()
nperm <- 999
for (i in  1:length(metavars)){
  # for (i in  1:2){
  # i <- 3
  mvar <- metavars[i]
  mvar2 <- mvar %>% gsub(' ','.',x = .) %>% gsub('/','.',x = .)
  print(mvar2)

  ####### 16s
  ######Rare 2k
  phyXsel <- phyXrare2k
  samdf <- data.frame(phyXsel@sam_data,demo = phyXsel@sam_data %>% data.frame() %>% dplyr::s
  phyXsel@sam_data <- sample_data(samdf)
  phyXsel <- phyXsel %>%
    subset_samples(!is.na(demo)) %>%
    transform_sample_counts( function(x) x  / sum(x))
  D16s <- phyXsel %>% phyloseq::distance("bray")

  dfsel <- data.frame(sample_data(phyXsel))
  dfselun <- dfsel%>% filter(!duplicated(Participant))

  r16s <- adonis2(D16s ~ demo, data = dfsel, permutations = 0) %>% tidy %>%
    mutate(demo = mvar, omics = 'ME-16s')

  r16sperm <- data.frame()
  for (j in 1:nperm){
    dfselperm <- dfsel %>%
      left_join(dfselun %>% mutate(demo2 = sample(demo)), by = 'Participant')
    r16sperm <- r16sperm %>%
      bind_rows(adonis2(D16s ~ demo2, data = dfselperm, permutations = 0) %>% tidy %>%
                  mutate(demo = mvar, omics = 'ME-16s') %>% filter(term=='Model'))
  }


  pv <- sum(r16sperm$statistic>r16s$statistic[1]) / dim(r16sperm)[1]
  r16s$p.value2 <- NA
  r16s$p.value2[1] <- pv
  r16srare2k <- r16s
```

```
######Rare 10k
phyXsel <- phyXrare10k
samdf <- data.frame(phyXsel@sam_data,demo = phyXsel@sam_data %>% data.frame() %>% dplyr::s
phyXsel@sam_data <- sample_data(samdf)
phyXsel <- phyXsel %>%
  subset_samples(!is.na(demo)) %>%
  transform_sample_counts( function(x) x  / sum(x))
D16s <- phyXsel %>% phyloseq::distance("bray")

dfsel <- data.frame(sample_data(phyXsel))
dfselun <- dfsel%>% filter(!duplicated(Participant))

r16s <- adonis2(D16s ~ demo, data = dfsel, permutations = 0) %>% tidy %>%
  mutate(demo = mvar, omics = 'ME-16s')

r16sperm <- data.frame()
for (j in 1:nperm){
  dfselperm <- dfsel %>%
    left_join(dfselun %>% mutate(demo2 = sample(demo)), by = 'Participant')
  r16sperm <- r16sperm %>%
    bind_rows(adonis2(D16s ~ demo2, data = dfselperm, permutations = 0) %>% tidy %>%
                mutate(demo = mvar, omics = 'ME-16s') %>% filter(term=='Model'))
}

pv <- sum(r16sperm$statistic>r16s$statistic[1]) / dim(r16sperm)[1]
r16s$p.value2 <- NA
r16s$p.value2[1] <- pv
r16srare10k <- r16s



##### ITS - 00reads
phyXsel <- phyITS_00
samdf <- data.frame(phyXsel@sam_data,demo = phyXsel@sam_data %>% data.frame() %>% dplyr::s
phyXsel@sam_data <- sample_data(samdf)
phyXsel <- phyXsel %>%
  subset_samples(!is.na(demo)) %>%
  transform_sample_counts( function(x) as.numeric(x>0))
DITS <- phyXsel %>% phyloseq::distance("euclidean")

dfsel <- data.frame(sample_data(phyXsel))
dfselun <- dfsel%>% filter(!duplicated(Participant))
```

```
rITS <- adonis2(DITS ~ demo, data = dfsel, permutations = 0) %>% tidy %>%
  mutate(demo = mvar, omics = 'ME-ITS')

rITSperm <- data.frame()
for (j in 1:nperm){
  dfselperm <- dfsel %>%
    left_join(dfselun %>% mutate(demo2 = sample(demo)), by = 'Participant')
  rITSperm <- rITSperm %>%
    bind_rows(adonis2(DITS ~ demo2, data = dfselperm, permutations = 0) %>% tidy %>%
                mutate(demo = mvar, omics = 'ME-ITS') %>% filter(term=='Model'))
}

pv <- sum(rITSperm$statistic>rITS$statistic[1]) / dim(rITSperm)[1]
rITS$p.value2 <- NA
rITS$p.value2[1] <- pv
rITS_00 <- rITS


##### ITS - 100 reads
phyXsel <- phyITS_100
samdf <- data.frame(phyXsel@sam_data,demo = phyXsel@sam_data %>% data.frame() %>% dplyr::s
phyXsel@sam_data <- sample_data(samdf)
phyXsel <- phyXsel %>%
  subset_samples(!is.na(demo)) %>%
  transform_sample_counts( function(x) as.numeric(x>0))
DITS <- phyXsel %>% phyloseq::distance("euclidean")

dfsel <- data.frame(sample_data(phyXsel))
dfselun <- dfsel%>% filter(!duplicated(Participant))

rITS <- adonis2(DITS ~ demo, data = dfsel, permutations = 0) %>% tidy %>%
  mutate(demo = mvar, omics = 'ME-ITS')

rITSperm <- data.frame()
for (j in 1:nperm){
  dfselperm <- dfsel %>%
    left_join(dfselun %>% mutate(demo2 = sample(demo)), by = 'Participant')
  rITSperm <- rITSperm %>%
    bind_rows(adonis2(DITS ~ demo2, data = dfselperm, permutations = 0) %>% tidy %>%
                mutate(demo = mvar, omics = 'ME-ITS') %>% filter(term=='Model'))
}
```

```
  pv <- sum(rITSperm$statistic>rITS$statistic[1]) / dim(rITSperm)[1]
  rITS$p.value2 <- NA
  rITS$p.value2[1] <- pv
  rITS_100 <- rITS



  resDEMO_stab <- bind_rows(resDEMO_stab,
                            r16srare2k %>% mutate(analysis = 'rare2k'),
                            r16srare10k %>% mutate(analysis = 'rare10k'),
                            rITS_00 %>% mutate(analysis = '>0'),
                            rITS_100 %>% mutate(analysis = '>100'))

}

save(file = 'betademo_res_stab.RData', list = c('resDEMO_stab','metavars'))
```

Overall explained variance versus demo

Figure 4A in publication

```
fmrt <- list(
  R2 = colDef(format = colFormat(percent = TRUE, digits = 2)),
  p.value2 = colDef(name = 'p-value', format = colFormat(digits = 3))
)

resDEMO %>%
  filter(term=='Model' | term=='val') %>%
  filter(!is.na(df)) %>%
  filter(!str_detect(omics,'sqrt')) %>%
  dplyr::select(demo,omics,df,R2,p.value2) %>%
  reactable(columns = fmrt) %>% add_subtitle('Overall variance explained (on beta-diversity)
```

Overall variance explained (on beta-diversity) by individual demographic information

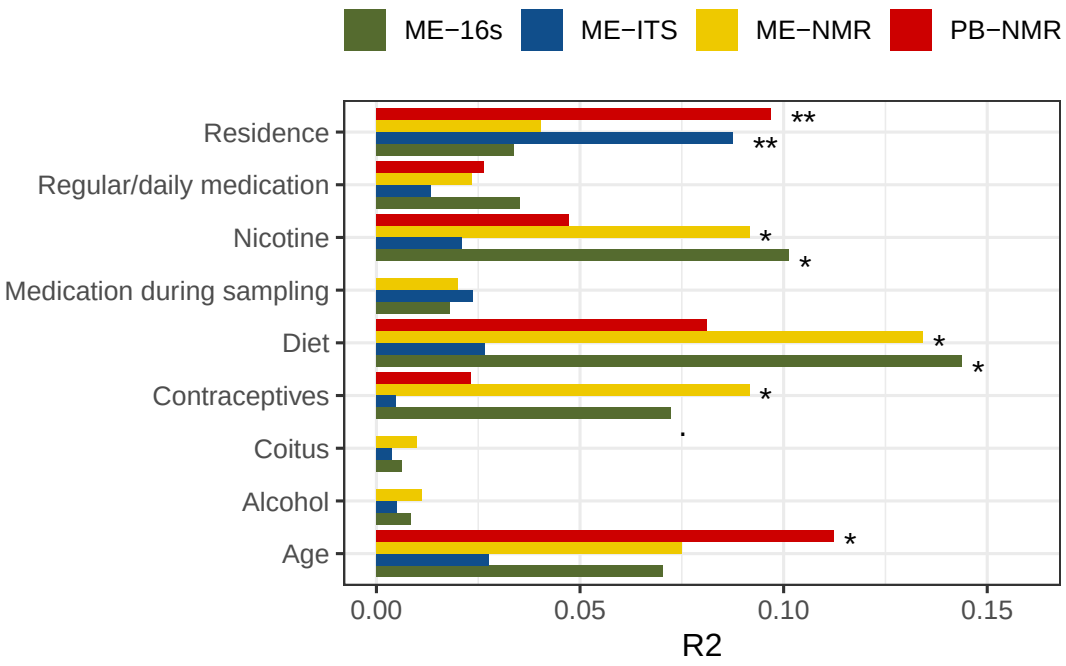
demo	omics	df	R2	p-value
Age	ME-NMR	2	7.49%	0.273
Age	PB-NMR	2	11.25%	0.044
Age	ME-16s	2	7.04%	0.528
Age	ME-ITS	2	2.77%	0.393
Residence	ME-NMR	1	4.03%	0.238
Residence	PB-NMR	1	9.69%	0.003
Residence	ME-16s	1	3.37%	0.497
Residence	ME-ITS	1	8.76%	0.001
Diet	ME-NMR	2	13.43%	0.028
Diet	PB-NMR	2	8.12%	0.248

1–10 of 33 rows Previous **1** 2 3 4 Next

```
resDEMO %>%
  filter(term=='Model' | term=='val') %>%
  filter(!str_detect(omics,'sqrt')) %>%
  rio::export(file = 'Figure4A_data.xlsx')

Fig4A <- resDEMO %>%
  mutate(pv = myround2(p.value2),
         pv2 = gtools::stars.pval(p.value2)) %>%
  filter(term=='Model' | term=='val') %>%
  filter(!str_detect(omics,'sqrt')) %>%
  ggplot(data = ., aes(demo,R2, fill = omics, label = pv2)) +
  geom_bar(stat = 'identity', position = position_dodge()) +
  coord_flip() +
  ylim(c(0,0.16)) +
  geom_text(position = position_dodge(width = 1), hjust = -0.8, vjust = 0.9) +
  scale_fill_manual(values = colorcodes) +
  theme_bw() +
  labs(fill = NULL) +
  theme(legend.position = 'top') +
```

```
  xlab(NULL)
Fig4A
```

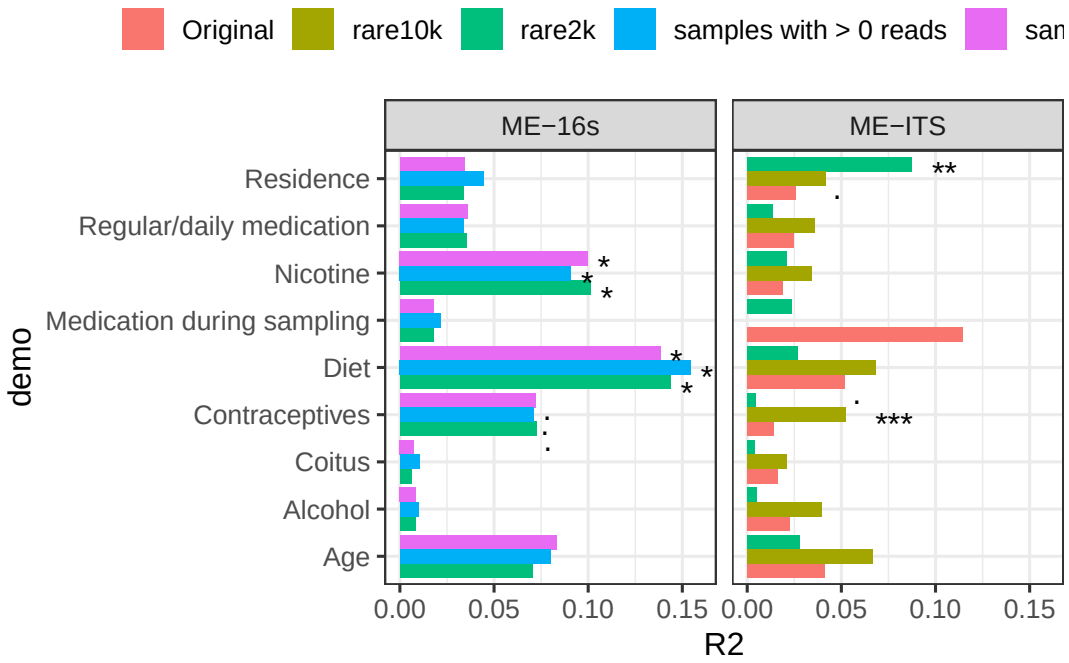



```
ggsave(filename = 'FIG4_betadiv.pdf',  plot = Fig4A,width = 6,height = 5.2)
```

Statbility of results

```
resDEMO %>% mutate(analysis = 'original') %>% bind_rows(resDEMO_stab) %>%
  mutate(pv = myround2(p.value2),
         pv2 = gtools::stars.pval(p.value2)) %>%
  filter(term=='Model') %>%
  filter(!str_detect(omics,'sqrt')) %>%
  filter(omics=='ME-16s' |omics=='ME-ITS') %>%
  ungroup %>%
  mutate(analysis = analysis %>% factor(labels = c('Original','rare10k','rare2k','samples wi
  ggplot(data = ., aes(demo,R2, fill = analysis, label = pv2)) +
  geom_bar(stat = 'identity', position = position_dodge()) +
  coord_flip() +
  ylim(c(0,0.16)) +
  geom_text(position = position_dodge(width = 1), hjust = -0.8, vjust = 0.9) +
  # scale_fill_manual(values = colorcodes) +
  theme_bw() +
```

```
  labs(fill = NULL) +
  theme(legend.position = 'top') +
  facet_grid(.~omics)
```




```
  xlab(NULL)
```

```
<ggplot2::labels> List of 1
 $ x: NULL
```

Examples

Figure 4B in publication

```
dfsel1 <- MEconc_newlong %>%
  ungroup %>%
  filter(str_detect(met,'unk29-asparagine') |
           str_detect(met,'arabinose')) %>% # |
  filter(!is.na(Diet))  %>%
  dplyr::select(met,Participant,Diet,Cycle,`Cycle day`,logy)

dfsel2 <- MEconc_newlong %>%
```

```
  ungroup %>%
  filter(!is.na(Nicotine)) %>%
  filter(str_detect(met,'trimethylamine')) %>%
  dplyr::select(met,Participant,Nicotine,Cycle,`Cycle day`,logy)

dfsel1 %>% rio::export(file = 'Figure4B1_data.xlsx')
dfsel2 %>% rio::export(file = 'Figure4B2_data.xlsx')

g1 <- MEconc_newlong %>%
  filter(str_detect(met,'unk29-asparagine') |
           str_detect(met,'arabinose')) %>% # |
  # str_detect(met,'bin-ornithine-creatinine-tyrosine')) %>%
  mutate(met = met %>% gsub('unk29-','',x = .) %>% gsub('bin-','',x = .)) %>%
  filter(!is.na(Diet)) %>%
  ggplot(data = ., aes(Diet,logy, fill = Diet)) +
  geom_boxplot() +
  scale_fill_manual(values = c('red','blue','chartreuse3')) +
  ggbeeswarm::geom_beeswarm() +
  theme_bw() +
  # facet_wrap(~met, scales = 'free', ncol = 1) +
  facet_grid(met~., scales = 'free') +
  theme(legend.position = 'none')

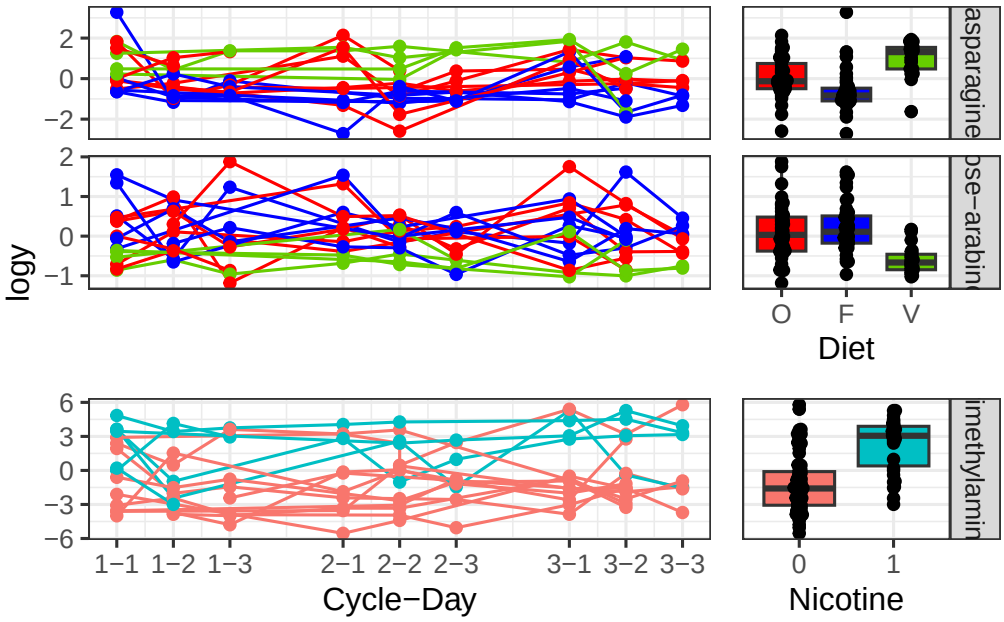
g1b <- MEconc_newlong %>%
  mutate(Nicotine = Nicotine %>% factor()) %>%
  filter(str_detect(met,'trimethylamine')) %>%
  filter(!is.na(Nicotine)) %>%
  ggplot(data = ., aes(Nicotine,logy, fill = Nicotine)) +
  geom_boxplot() +
  ggbeeswarm::geom_beeswarm() +
  theme_bw() +
  # facet_wrap(~met, scales = 'free', ncol = 1) +
  facet_grid(met~., scales = 'free') +
  theme(legend.position = 'none')

g2 <- MEconc_newlong %>%
  filter(str_detect(met,'unk29-asparagine') |
           str_detect(met,'arabinose')) %>% # |
  mutate(met = met %>% gsub('unk29-','',x = .) %>% gsub('bin-','',x = .)) %>%
  mutate(met = met %>% gsub('unk','',x = .)) %>%
  filter(!is.na(Diet)) %>%
  mutate(cycle_day = parse_number(Cycle)*4 + `Cycle day`) %>%
```

```
  ggplot(data = ., aes(cycle_day,logy, group = Participant, color =Diet)) +
  geom_line() + geom_point() +
  scale_x_continuous(breaks = c(5,6,7,9,10,11,13,14,15),
                     labels =  c('1-1','1-2','1-3','2-1','2-2','2-3','3-1','3-2','3-3')) +
  scale_color_manual(values = c('red','blue','chartreuse3')) +
  # facet_wrap(~met, scales = 'free', ncol = 1) +
  facet_grid(met~., scales = 'free') +
  theme_bw() +
  xlab('Cycle-Day') +
  theme(legend.position = 'none',
    strip.text.y       = element_blank(),
    strip.background.y = element_blank()
  )

g2b <- MEconc_newlong %>%
  filter(str_detect(met,'trimethylamine')) %>%
  filter(!is.na(Nicotine)) %>%
  mutate(cycle_day = parse_number(Cycle)*4 + `Cycle day`,
         Nicotine = Nicotine %>% factor()) %>%
  ggplot(data = ., aes(cycle_day,logy, group = Participant, color =Nicotine)) +
  geom_line() + geom_point() +
  scale_x_continuous(breaks = c(5,6,7,9,10,11,13,14,15),
                     labels =  c('1-1','1-2','1-3','2-1','2-2','2-3','3-1','3-2','3-3')) +
  # facet_wrap(~met, scales = 'free', ncol = 1) +
  facet_grid(met~., scales = 'free') +
  theme_bw() +
  xlab('Cycle-Day') +
  theme(legend.position = 'none',
    strip.text.y       = element_blank(),
    strip.background.y = element_blank()
  )

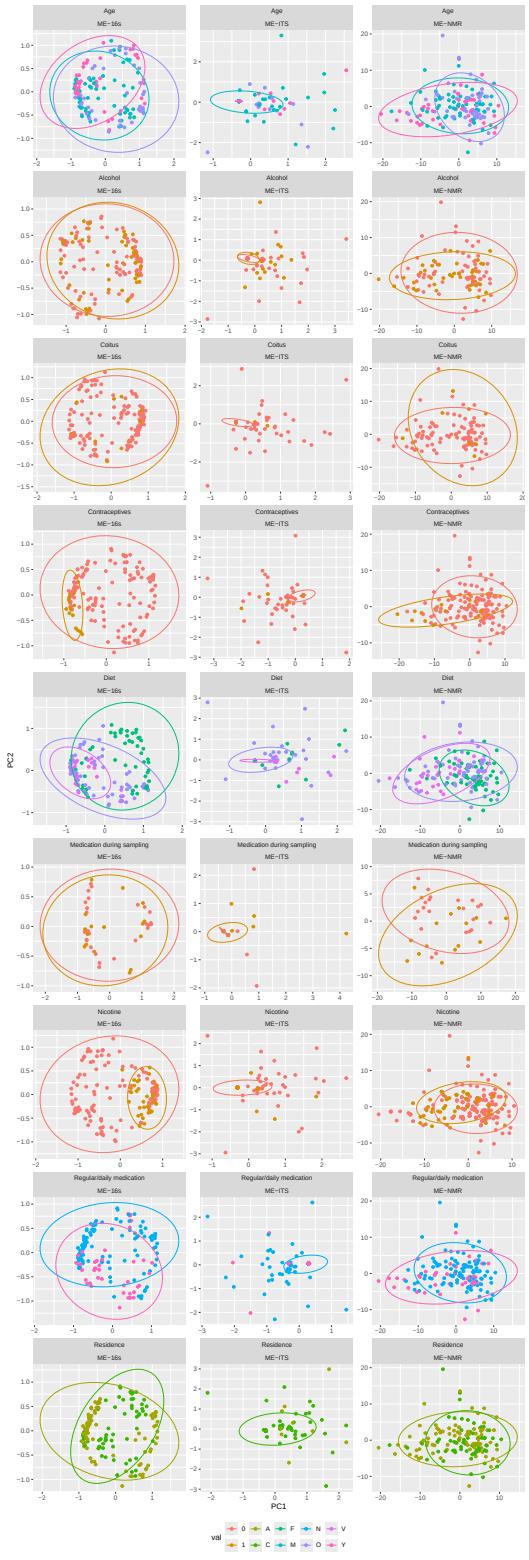
layout <- "
AAAB
AAAB
CCCD
"
FIG2b <- g2 + g1 + g2b + g1b + plot_layout(design = layout , axes = 'collect')
FIG2b
```




```
ggsave(filename = 'FIG4_dietexamples.pdf',  plot = FIG2b,width = 5,height = 5)
```

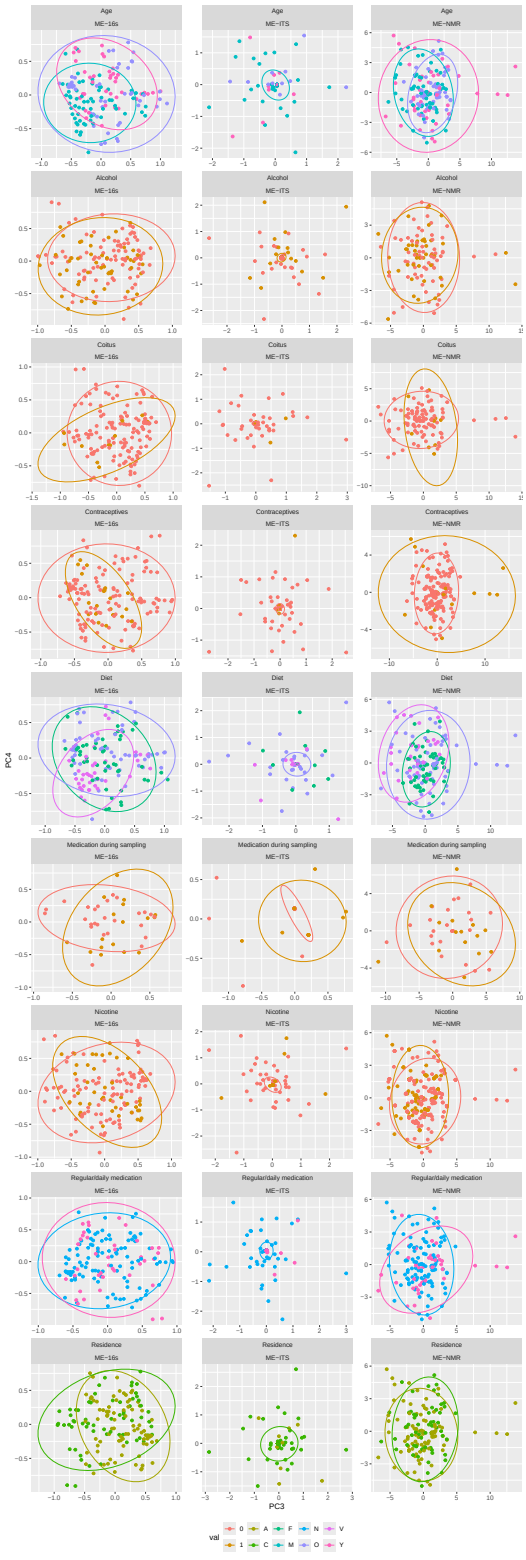
Multivariate distribution (PC1 vs PC2)

```
SCORESdemo %>%
  ggplot(data = ., aes(PC1,PC2, color = val)) +
  geom_point() +
  stat_ellipse(aes(group = val)) +
  facet_wrap(~demo + omics, scales = 'free', ncol = 3) +
  theme(legend.position = 'bottom')
```


Age
ME−16s
ME−ITS
ME−NMR
Alcohol
Coitus
Contraceptives
Diet
Medication during sampling
Nicotine
Regular/daily medication
Residence
PC1
PC2
val
0
1
A
C
F
M
N
O
V
Y

Multivariate distribution (PC3 vs PC4)

```
SCORESdemo %>%
  ggplot(data = ., aes(PC3,PC4, color = val)) +
  geom_point() +
  stat_ellipse(aes(group = val)) +
  facet_wrap(~demo + omics, scales = 'free', ncol = 3) +
  theme(legend.position = 'bottom')
```


Age
Alcohol
Coitus
Contraceptives
Diet
Medication during sampling
Nicotine
Regular/daily medication
Residence
ME−16s
ME−ITS
ME−NMR
PC4
PC3
val
0
1
A
C
F
M
N
O
V
Y